A 3.6mW 5MS/s 100dB-SNDR 2-Step SAR ADC with Continuous-Time Second-Stage and Tracking Averaging

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***Abstract*—The residue amplifier (RA) often dominates the noise and power in a high-precision 2-step SAR ADC. In this work, we propose to use the continuous-time (CT) SAR as the second stage, eliminating the sampling operation after RA and thus preventing the RA noise from aliasing into the in-band. Meanwhile, combining with the tracking averaging based LSB repeating technique, the wideband RA noise is reduced by 7dB.The prototype ADC implemented in 180nm CMOS measures ~100dB SNDR with 5MS/s sampling rate. The power consumption is 3.6mW, leading to a Schreier FoM of 187.8dB with 100kHz input frequency.**

***Keywords*—2-step SAR, CT SAR, LSB repeating, ring amplifier**

I. Introduction

The SAR ADCs has demonstrated excellent energy efficiency, owing to their simple structure and friendliness to process scaling. However, a typical SAR ADC usually has limited precision of ≤12-bit ENOB due to comparator noise. To reduce noise, several LSB repeating techniques are developed [1-3]. By which, the comparator noise can be averaged by multiple comparisons. However, the noise reduction effect by LSB repeating is limited, the ENOB improvement is usually below 1 bit. The 2-step SAR ADCs, which suppress the comparator noise by introducing a residue amplifier (RA), can boost the ENOB of ADC to the level of 16 bit [4-7]. Nevertheless, the design of a low-noise and low-power RA is challenging.

This work aims to realize a high-precision 2-step SAR ADC and further enhances its energy efficiency. By using a continuous-time (CT) SAR ADC in the second step and incorporating with a LSB repeating technique, the noise from both RA and comparator can be significantly reduced. A prototype ADC is fabricated in 180nm process. Operating with 5MS/s sampling rate, it measures ~100dB SNDR with input frequencies up to 100kHz. The power consumption is 3.6mW, leading to the Schreier FoM (FoMs) of 187.8dB. This work achieves the best FoMs among Nyquist ADCs published to date.

II. Proposed Architecture

Figure 1 illustrates the principle of proposed 2-step SAR ADC. For a conventional 2-step SAR ADC, the second stage is a DT one. The RA noise will be aliased into in-band (fs/2) due to sampling. To reduce noise, the large capacitor size in second stage has to be used, which is at the cost of increased power consumption and reduced operation speed. To address this issue, this work proposes to realize the second stage using a CT SAR ADC [8] with LSB repeating. M times of LSB repeating is equivalent to M times of oversampling, thus the RA noise will be aliased into a higher frequency band of M·fs/2. The aliased noise will be further filtered by a digital downsampling filter, thereby substantially reducing the RA noise. Together with RA noise, the comparator noise and quantization error can also be suppressed by the LSB repeating.

A tracking-averaging [1] based LSB repeating technique is used in this work. As shown in Figure 1, V_{RES} is the residue voltage on DAC_2 before LSB repeating, D_T is the comparison result during LSB repeating. The tracking is basically realized by feeding back the first-

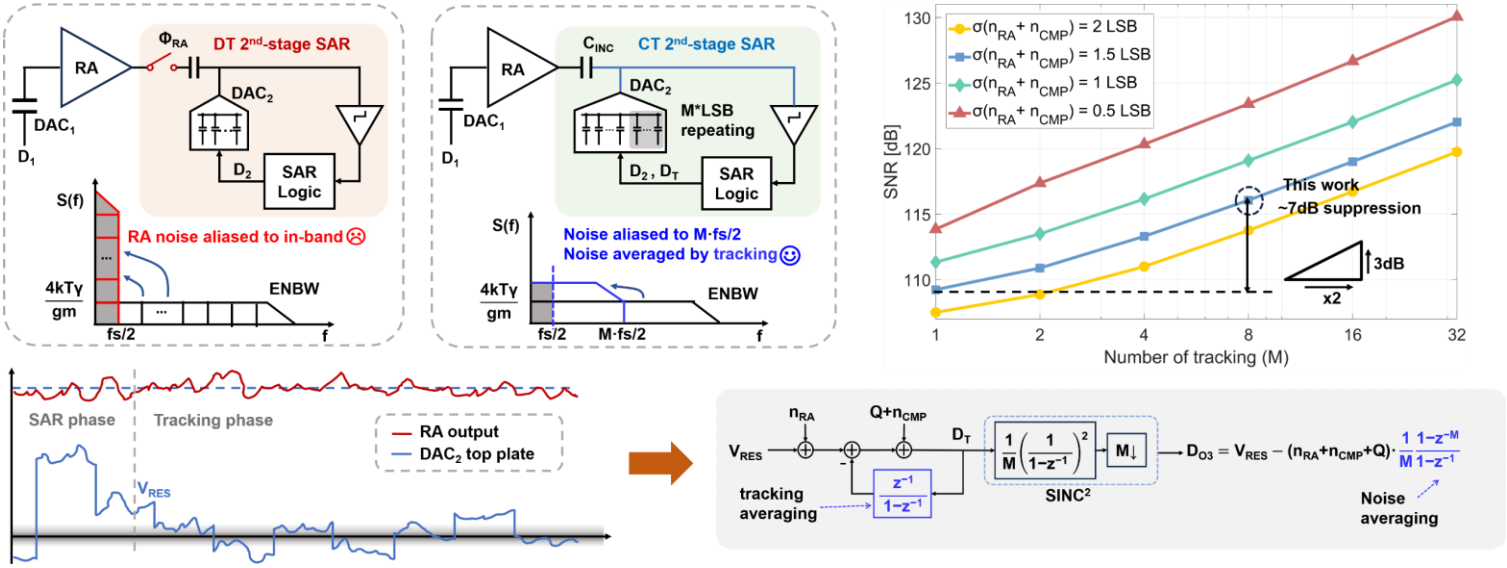


Fig. 1. Principle of proposed 2-step SAR ADC with CT second stage and tracking averaging.

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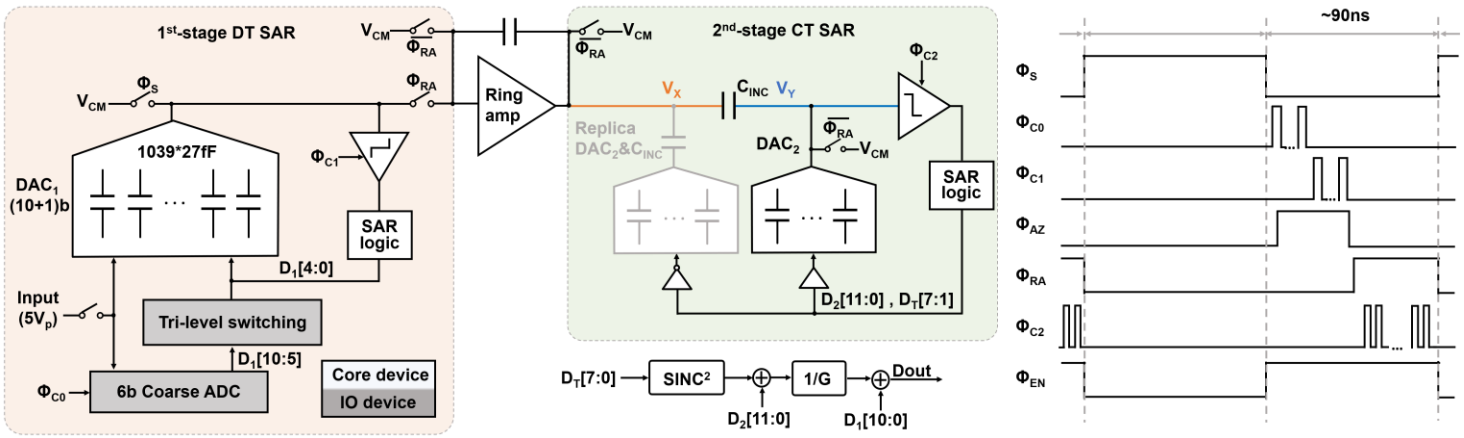


Fig. 2. The proposed ADC architecture and timing diagram

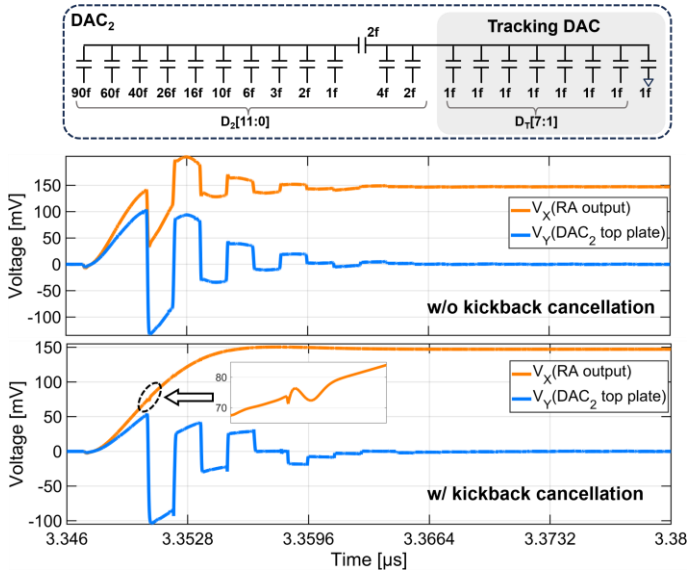


Fig. 3. The implementation of DAC_2 and effect of kickback cancellation.

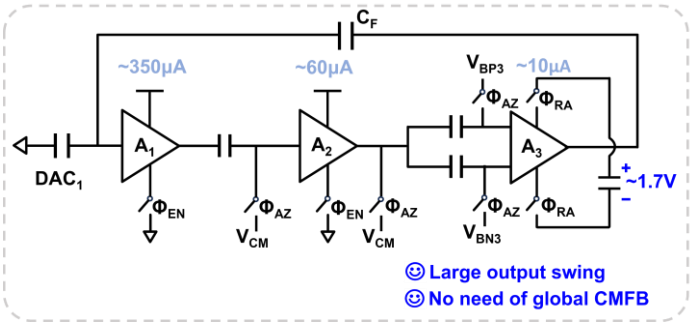


Fig. 4. The implementation of ring amplifier.

order integration results of D_T. Different with previous works, we employ a $SINC^2$ digital filter to process D_T. Theoretically, after digital filtering, the RA noise, comparator noise and quantization error can be averaged by M times.

The behavioral modeling is conducted to verify the proposed method. It shows that the SNR increases by ~3dB with every twofold increase of M. With low RA and comparator noise (σ=0.5LSB), 8 repeatings can achieve 9dB noise reduction. However, when the noise level is high, the noise suppression effect is slightly diminished. This is because at high noise levels, the residual voltage at end of SAR phase may deviate significantly from the comparator threshold voltage, requiring additional few cycles to pull it back. Consequently, the effective number of LSB repeating is reduced. In this design, the noise of RA and comparator is designed to be about σ=1.5LSB, leading to ~7dB noise reduction.

III. Circuit Implementation

A. ADC Architecture

Figure 2. illustrates the architecture of the proposed ADC. It consists of a 10-bit DT SAR as the first stage and a CT SAR with tracking averaging as the second stage. In the second stage, the CT SAR contributes 11-bit resolution in itself, while another 1.5-bit resolution is obtained by 8 times of LSB repeating. The 2-bit inter-stage redundancy is provided to tolerate the offset and noise from the first-stage conversion, leading to an overall ADC resolution of 20.5 bit. The tracking is realized by simply adding 7 unit capacitors to the DAC of second stage (DAC_2) and feeding back the comparison results D_T[7:1] to them sequentially.

In the first stage, a coarse ADC is used to resolve the 6-bit MSB. The MSB result is then applied to DAC_1 through a tri-level switching [7], reducing the reference power by half. The coarse ADC and the bottom-plate switches of DAC_1 are realized with IO devices under a 5V supply to support a $5V_P$ input swing, while all the other circuits are implemented with core devices under a 1.8V supply to reduce power. A ringamp [9] is used as the RA, which provides 32× inter-stage gain. The total sampling capacitor is 28pF to meet kT/C noise requirement.

B. Continuous-Time 2nd-Stage SAR

To realize the CT SAR, the RA output (V_X) is coupled to the top plate of DAC_2 (V_Y) through a capacitor C_{INC}. V_Y is connected to V_{CM} when Φ_{RA} is low, defining its DC voltage. However, Vx can be susceptible to kickback caused by the switching activities of DAC_2. Because the RA has limited output impedance, the voltage jumps at V_Y during SAR conversion can kick back to V_X through C_{INC}, as shown in Figure 3. As a result, both V_X and V_Y are affected, leading to the wrong conversion results. To address this issue, we propose a kickback cancellation technique by introducing a replica C_{INC} in series with a replica DAC_2. The replica C_{INC} and DAC_2 generate the opposite kickback by feeding back the opposite digital outputs, and the delays on both paths are carefully matched, thereby canceling the original

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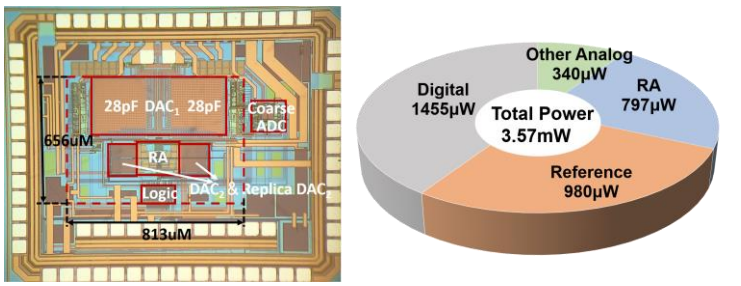


Fig. 5. Chip micrograph and power breakdown.

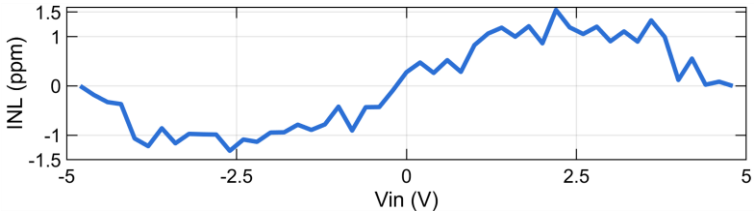


Fig. 6. Measured ADC INL.

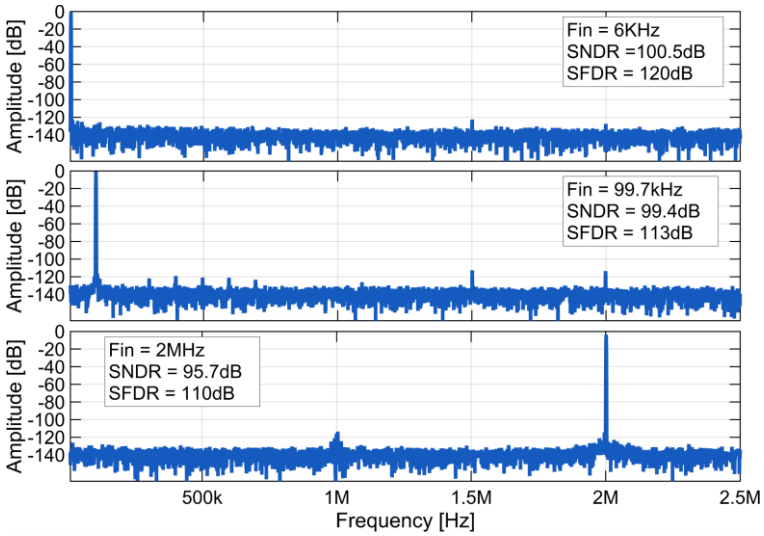


Fig. 7. Measured ADC output spectra with input frequency of 6kHz, 100kHz and 2MHz (20k-point FFT).

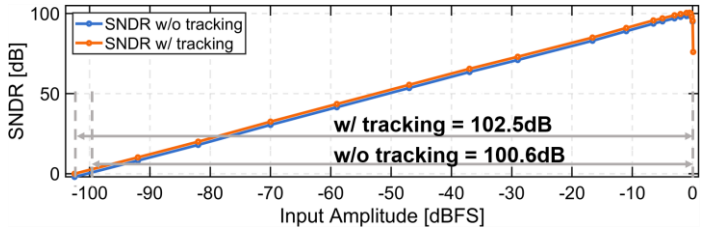


Fig. 8. Measured dynamic range.

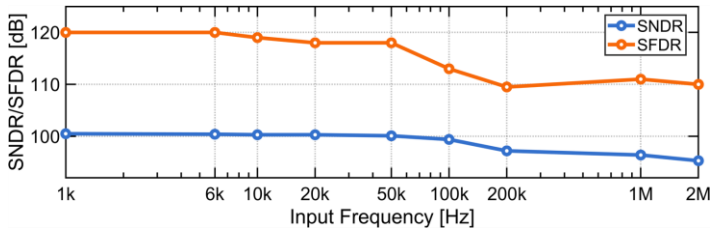


Fig. 9. Measured SNDR and SFDR versus input frequency.

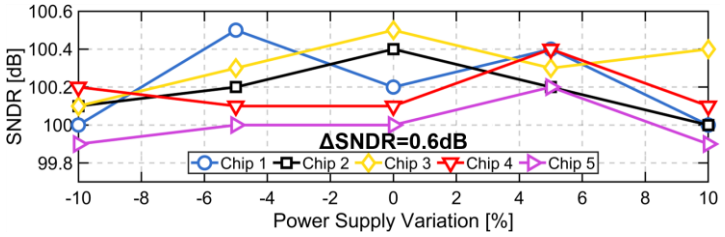


Fig. 10. Measured SNDR versus power supply variation across 5 samples.

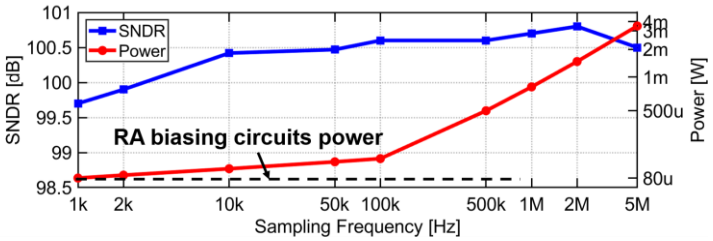


Fig. 11. Measured SNDR and power consumption versus sampling frequency.

kickback at V_X node. The additional power and area introduced by the replica DAC_2 are negligible, since the capacitor size in the second stage is small. To tolerate the RA setting error during the initial phase of second-stage conversion, the substantial redundancy is added in DAC_2.

The LSB repeating only consumes 5% (10ns) of an entire ADC cycle, therefore the penalty on and speed is minimal. While it reduces the RA and comparator noise by 7dB and offers 2dB SNR improvement to the overall ADC, making it worthwhile.

C. Ringamp Design

The implementation of ringamp [9] for RA is shown in Figure 4. The first and second stage of RA are static CMOS-input amplifier with local common-mode feedback (CMFB). Owing to the noise suppression by tracking averaging, the first stage of RA only consumes 350μA current, which is significantly lower than other designs of the same type [6-7]. The output stage uses the floating inverter amplifier (FIA) [10]. FIA exhibits good common-mode stability, thus no CMFB is required. In this manner, every stage has the stable common mode in local, and the RA as a whole is free from complex global CMFB. The input transistors of FIA are biased in subthreshold region. The voltage on C_{RES} drops slightly during residue amplification, avoiding the excessive degradation of RA output swing. Auto-zero (AZ) is implemented between A_1 and A_2 to suppress offset and 1/f noise. AZ is performed during the conversion of first-stage SAR. A_1 and A_2 are turned off during ADC sampling phase to reduce power.

IV. Measurement Results

The prototype 2-step SAR ADC is fabricated in 180nm process. As shown in Figure 5, the core area is 0.53mm^2. Operating with 5MS/s sampling rate, it consumes 3.57mW power. The power breakdown is 0.8mW for RA, 0.98mW for reference, 1.45mW for digital circuits and 0.34mW for other analog circuits. The off-chip LMS-based foreground calibration is used to address DAC mismatch and RA gain error. Figure 6 shows the measured INL. The maximum INL is 1.6ppm. Figure 7 shows the measured output spectra. The SNDR is 100.5dB with 6kHz input, 99.4dB with 100kHz input and 95.7dB with 2MHz input, respectively. Figure 8 shows dynamic range measurement result. Without tracking averaging, the measured DR is 100.6dB. With tracking averaging, the DR is improved to 102.5dB. Figure 9 provides

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TABLE I. PERFORMANCE SUMMARY AND COMPARISON

	This work	ISSCC 22 Steensgard	ISSCC 24 Bodnar	JSSC 25 Huang	VLSI 14 Bannon	ISSCC 19 ElShater	AD4020 [11]	LT2378 [12]
Architecture	2-Step SAR	Pipe SAR	TI-Pipe SAR	SAR	Pipe SAR	Pipe SAR	SAR	SAR
Noise/power reduction technique?	CT 2nd stage with tracking averaging	Incomplete settling RA	2-phase settling RA	SRM	-	-	-	-
Process [nm]	180	180	40	180	180	180	-	-
Fs [MS/s]	5	2	40	5	5	15	1.8	1
Resolution [bit]	20.5	24	20	16	18	16	20	20
Area [mm²]	0.53	0.78	0.9	0.57	5.74	1.82	-	-
Supply [V]	5 /1.8	5 /1.8	3.3 /1.1	3.3 /1.8	5 /1.8	3.3 /1.8	5 /1.8	5 /2.5
C_S [pF]	28	50	20.5	21	-	17	40	45
Power [mW]	3.6	8.5	50	5.3	30.5	10	15	21
DR [dB]	102.5	106.3	94.2	95	100.2	95	101	-
SNDR [dB] @Fin [Hz]	100.5@6k 99.4@100k 95.7@2M	105.3@100k	93.7@1k 92@1M	93.7@2k 89.1@1M	98.6@1k	91@50k 88@4M	96.5@100k	102@100k
FoM_S [dB] @Fin [Hz]	187.8@100k 184.1@2M	186.0@100k	179.7@1k 178@1M	180.4@2k 175.8@1M	177.7@1k	179.8@50k 176.8@4M	174.3@100k	175.8@100k

$FoM_S = SNDR + 10*\lg(\frac{BW}{Power})$

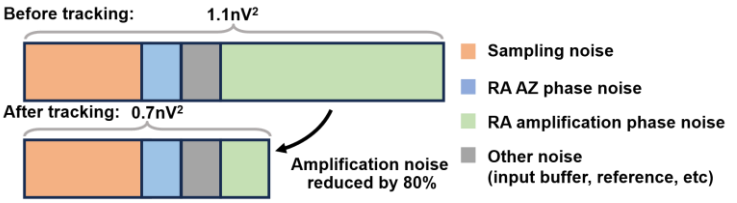


Fig. 12. Noise breakdown before and after tracking.

measured performance with sweeping input frequency. The measured SNDR is around 100dB with input frequency within 100kHz. Limited by signal source and input driver, a notable SNDR drop can be observed when input frequency is higher than 100kHz. To prove robustness, Figure 10 provides 5 chip samples measurement results over ±10% analog supply variation. The measured SNDR variation is less than 0.6dB. Figure 11 shows the measured results with sampling frequency scaling from 1kS/s to 5MS/s under 1kHz input frequency, the SNDR maintains around 100dB, and the power scaling is limited by RA bias circuit of 80uW. Figure 12 shows the noise breakdown, with RA amplification noise reduced by 80% after tracking averaging, the total noise of ADC is reduced by 36%. Table I summaries the ADC performance and compares it with other high-precision SAR ADCs. These works have similar supply voltage and sampling capacitance. Among them, it has the highest SNDR with >1MHz input frequency. It achieves 187.8dB FoM_S with 100kHz input, and 184.3dB FoM_S with 2MHz input.

V. CONCLUSION

A high-precision 2-step SAR ADC is presented in this work. It uses a CT SAR with tracking averaging as the 2nd stage, reducing the RA noise by 80%, leading to 2dB SNR improvement with minimal power and time penalty. Owing to these techniques and careful design, this work achieves the state-of-the-art performance and energy efficiency.

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